

Pathology reports

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Pathology Report DB

Pathology data

환자의 병리결과지를 텍스트로 선택하여 “*.txt” 파일로 저장함. 병리결과지를 저장하는 방법이 여러가지 형태가 있는데 병리와 데이터베이스에 직접 접속하지 않으면서 작업하려면 병리결과지를 Copy & Paste 형태로 메모장 프로그램에 옮겨오고, 차트번호를 파일이름으로 하여 저장하는 것이 현재로서는 가장 간단한 방법일 것 같다.

“P00000000.txt” 파일이름으로 한개의 디렉토리에 전부 저장

```
library(stringr)
library(magrittr)
getwd()
```

```
[1] "G:/R project/Metastasectomy"
```

```
dir(pattern = "*.txt")
```

```
[1] "P00862499.txt" "P00883814.txt" "P00907260.txt" "P00912360.txt"
[5] "P00964294.txt" "P00989382.txt" "P01086311.txt" "P01107342.txt"
[9] "P01117777.txt" "P01117868.txt" "P01118305.txt" "P01122443.txt"
[13] "P01129937.txt" "P01130720.txt" "P01153426.txt" "P01189766.txt"
[17] "P01190449.txt" "P01192548.txt" "P01192606.txt" "P01192791.txt"
[21] "P01193764.txt" "P01203731.txt" "P01239418.txt" "P01252315.txt"
[25] "P01253357.txt" "P01274950.txt" "P01312805.txt" "P01327517.txt"
[29] "P01335041.txt" "P01337742.txt" "P01340266.txt" "P01341373.txt"
[33] "P01344205.txt" "P01345120.txt" "P01348893.txt" "P01352361.txt"
[37] "P01352706.txt" "P01353586.txt" "P01353782.txt" "P01354875.txt"
[41] "P01367068.txt" "P01368869.txt"
```

```
dir(pattern = "*.txt") -> txtfiles
```

txtfiles에 파일이름을 전부 저장 - 파일이름이 병록번호이므로 chartno 변수에 전부 저장
chartno와 report 라는 변수를 가진 data.frame을 만들 계획

```
txtfiles %>% str_match(pattern = "\\d{7,8}")
```

```
      [,1]  
[1,] "00862499"  
[2,] "00883814"  
[3,] "00907260"  
[4,] "00912360"  
[5,] "00964294"  
[6,] "00989382"  
[7,] "01086311"  
[8,] "01107342"  
[9,] "01117777"  
[10,] "01117868"  
[11,] "01118305"  
[12,] "01122443"  
[13,] "01129937"  
[14,] "01130720"  
[15,] "01153426"  
[16,] "01189766"  
[17,] "01190449"  
[18,] "01192548"  
[19,] "01192606"  
[20,] "01192791"  
[21,] "01193764"  
[22,] "01203731"  
[23,] "01239418"  
[24,] "01252315"  
[25,] "01253357"  
[26,] "01274950"  
[27,] "01312805"  
[28,] "01327517"  
[29,] "01335041"  
[30,] "01337742"  
[31,] "01340266"  
[32,] "01341373"  
[33,] "01344205"  
[34,] "01345120"  
[35,] "01348893"  
[36,] "01352361"  
[37,] "01352706"  
[38,] "01353586"  
[39,] "01353782"  
[40,] "01354875"  
[41,] "01367068"  
[42,] "01368869"
```

```
txtfiles %>% str_match(pattern = "\\d{7,8}") -> chartno  
readLines(txtfiles[1]) %>% paste(sep=" ", collapse = "\n") %>% cat
```

Warning in readLines(txtfiles[1]): incomplete final line found on
'P00862499.txt'

【병리번호】S24-3784

【확 인 자】설혜실 / 설혜실

【 Gross 】

Left upper lobe. Received is a left upper lobectomy specimen, measuring 17x10x4cm. On serial section, a
<AB/CD/E/F/G; tumor, H; bronchial resection margin. I; remaining parenchyme>
Lymph node #7, #9, #10, #11. Entirely embedded.

** Frozen section diagnosis **

1. Bronchial resection margin ; No tumor involvement [고재수]
Received is a lump of fresh soft tissue, measuring 0.5x0.5cm. Entirely embedded.

【확인결과】

Lung, left upper lobe, lobectomy:

1. Invasive adenocarcinoma, acinar predominant
(acinar: 90%, lepidic: 10%), grade 2
with 1) size of invasion; 3.1x1.5cm
(total extent including lepidic pattern: 3.1x2cm)
2) no involvement of visceral pleura (PLO) (See NOTE)
3) lymphatic invasion: absent
4) vascular invasion: absent
5) perineural invasion: absent
6) spread through air space(STAS): absent
7) clear bronchial resection margin
[pathologic stage: pT2a]
2. No abnormal findings in the background lung

Lymph node, dissection:

No metastasis in 8 lymph nodes
(peribronchial; 0/1, #7: 0/1, #9: 0/2, #10: 0/2, #11: 0/2)
[pathologic stage: pN0]

<NOTE> 종양에 인접한 흉막이 이전 수술에 의한 유착 등으로 손상되어 있어 흉막 침윤여부를 정확하게 판단하기 어려우나,
평가가 가능한 부분에 한해서는 흉막 침윤이 관찰되지 않습니다.

<Proposed grading of resected early-stage invasive non-mucinous lung adenocarcinoma>

Grade 1: Lepidic-predominant with no or <20% high-grade pattern

Grade 2: Acinar or papillary-predominant with no or <20% high-grade pattern

Grade 3: Any tumor with 20% high-grade pattern (solid, micropapillary, cribriform, or complex glandular pa

*관련병리번호 :

SP24-262	Elastic Fiber Stain [E]	-	No visceral pleural invasion
SP24-263	Elastic Fiber Stain [F]	-	No visceral pleural invasion

[1] "【병리번호】S17-15025\n\n【확 인 자】박선후 / 고재수\n\n【 Gross 】\n\n#1: Right lung. Received is a right upper

```
chartno %>% . [22]
```

```
[1] "01203731"
```

```
pathology <- data.frame(chart.no= chartno, report = p.report)
pathology %>% str
```

```
'data.frame':  42 obs. of  2 variables:
 $ chart.no: chr  "00862499" "00883814" "00907260" "00912360" ...
 $ report  : chr  "\n[병리번호]S24-3784\n\n[확 인 자]설혜실 / 설혜실\n\n[ Gross  ]\n\n# Left upper lobe. Receive
```

report field에서 어떤 자료를 추출할 것인지 고려해야

병리번호, resection 방법

```
pattern01 <- "S\\d{2}-\\d{1,5}"
str_match_all(pathology$report, pattern = pattern01) %>% unlist
```

[1]	"S24-3784"	"S15-3730"	"S20-2581"	"S12-9100"	"S24-1433"	"S16-2211"
[7]	"S22-11161"	"S12-7517"	"S19-13203"	"S13-13939"	"S13-13839"	"S13-14039"
[13]	"S14-689"	"S14-6475"	"S24-7889"	"S15-5618"	"S17-1926"	"S17-2953"
[19]	"S17-4085"	"S17-4219"	"S17-4150"	"S17-4855"	"S17-15025"	"S19-13342"
[25]	"S19-13416"	"S19-13474"	"S24-3021"	"S24-10403"	"S24-10622"	"S24-171"
[31]	"S24-1960"	"S24-3367"	"S24-3412"	"S24-6606"	"S24-6512"	"S24-8154"
[37]	"S24-10382"	"S24-7268"	"S24-8536"	"S24-8771"	"S24-8910"	"S24-10982"
[43]	"S24-11532"					

```
pattern02 <- "확인결과[ ]{0,2}].*\\W{1,3}.* ((.*ectomy)|wedge resection)"
pattern02 %>% str_match(pathology$report, pattern = .) %>% unlist %>%
  .[,2]
```

[1]	"lobectomy"	"wedge resection"	"segmentectomy"	"lobectomy"
[5]	"lobectomy"	"wedge resection"	"lobectomy"	"pneumonectomy"
[9]	"lobectomy"	"pneumonectomy"	"lobectomy"	"lobectomy"
[13]	"lobectomy"	"lobectomy"	"lobectomy"	"pneumonectomy"
[17]	"segmentectomy"	"lobectomy"	"lobectomy"	"lobectomy"

[21]	"pneumonectomy"	"lobectomy"	"wedge resection"	"lobectomy"
[25]	"lobectomy"	"lobectomy"	"segmentectomy"	"lobectomy"
[29]	"lobectomy"	"lobectomy"	"segmentectomy"	"wedge resection"
[33]	"lobectomy"	"lobectomy"	"lobectomy"	"lobectomy"
[37]	"lobectomy"	"lobectomy"	"lobectomy"	"lobectomy"
[41]	"lobectomy"	"lobectomy"		

cell type

```
pattern03 <- "[\\w ]*carcinoma,?[\\w ]*"
pattern03 %>% str_match(pathology$report, pattern = .) %>% unlist
```

```
[,1]
[1,] " Invasive adenocarcinoma, acinar predominant"
[2,] " Adenocarcinom in situ vs Adenocarcinoma "
[3,] " Squamous cell carcinoma, poorly differentiated "
[4,] "      Invasive adenocarcinoma, acinar predominant "
[5,] " Invasive mucinous adenocarcinoma "
[6,] " Adenocarcinoma "
[7,] "      Squamous cell carcinoma, moderately differentiated"
[8,] "      Squamous cell carcinoma, moderately differentiated"
[9,] " Large cell carcinoma "
[10,] " Sarcomatoid carcinoma "
[11,] " Microinvasive adenocarcinoma, lepidic predominant "
[12,] "      Invasive adenocarcinoma, lepidic predominant "
[13,] " Invasive adenocarcinoma, lepidic predominant "
[14,] " Consistent with adenosquamous carcinoma"
[15,] "keratinizing squamous cell carcinoma, poorly differentiated"
[16,] "      Squamous cell carcinoma, moderately differentiated"
[17,] " Invasive adenocarcinoma, acinar predominant "
[18,] " Invasive adenocarcinoma, acinar predominant "
[19,] " Invasive adenocarcinoma, acinar predominant "
[20,] " Squamous cell carcinoma, moderately differentiated"
[21,] "      Squamous cell carcinoma, moderately differentiated"
[22,] " Squamous cell carcinoma, poorly differentiated"
[23,] " Adenocarcinoma "
[24,] "      Invasive adenocarcinoma, l papillary predominant "
[25,] " Pleomorphic carcinoma with adenocarcinoma"
[26,] " Squamous cell carcinoma, poorly differentiated"
[27,] "small cell carcinoma "
[28,] " Small cell carcinoma"
[29,] " Invasive adenocarcinoma, solid predominant"
[30,] " Squamous cell carcinoma, moderately differentiatede"
[31,] " Minimally invasive adenocarcinoma, nonmucinous "
[32,] " Adenocarcinoma "
[33,] "      Large cell carcinoma"
[34,] "mucinous lung adenocarcinoma"
[35,] "      Large cell neuroendocrine carcinoma"
```

```
[36,] " Invasive adenocarcinoma, papillary predominant"
[37,] " Invasive adenocarcinoma, acinar predominant "
[38,] " Adenocarcinoma "
[39,] " Invasive adenocarcinoma, acinar predominant "
[40,] " Invasive adenocarcinoma, micropapillary predominant"
[41,] " Squamous cell carcinoma, poorly differentiated"
[42,] " Squamous cell carcinoma, moderately differentiated"
```

size

```
pattern04 <- "size[~d][ ]?.*cm"
pattern04 %>% str_match(pathology$report, pattern = .) %>% unlist
```

```
[,1]
[1,] "size of invasion; 3.1x1.5cm"
[2,] "size; 0.6x0.6cm"
[3,] "size of invasion; 3.1x2.9x2.5cm"
[4,] "size of invasion; 5.5x3cm"
[5,] "size; 4.5x2.0x2.0cm"
[6,] "size of invasion; 0.3cm"
[7,] "size; 4.5x4.0cm"
[8,] "size; 4.2x4.1cm"
[9,] "size; 5.5x5.0x4.0cm"
[10,] "size; 6.8x6.0x5.0cm"
[11,] "size; 2.3x2.1cm"
[12,] "size; 2.6x2.5cm"
[13,] "size; 2.5x2.0cm"
[14,] "size of invasion; 6cm"
[15,] "size; 4.2x2.5x3.0cm"
[16,] "size; 2.0x1.0cm"
[17,] "size; 3.2x2.5cm"
[18,] "size; 1.7x1.3x1.2cm"
[19,] "size; 2.5x2.0cm"
[20,] "size; 2.5x2.4x1.5cm"
[21,] "size; 2.4x2.0cm"
[22,] "size of invasion; 4.4x1.8cm"
[23,] "size of invasion; 1.1x0.5cm"
[24,] "size of invasion; 3.7x3.8x3.5cm"
[25,] "size of invasion; 2.6x1.0cm"
[26,] "size of invasion; 5.5x5cm"
[27,] "size of invasion; 1.8x1.2cm"
[28,] "size of invasion; 2.5x2.5cm"
[29,] "size of invasion; 2.8x2.0x2.4cm"
[30,] "size of invasion; 8.7x7.0x5.5cm"
[31,] "size of invasion; 0.3cm"
[32,] "size of invasion; 1.3cm"
[33,] "size of invasion; 6x4x5cm"
[34,] "size of invasion; 3.0x3.0x2.6cm"
```

```
[35,] "size of invasion; 2.6x2.4cm"
[36,] "size of invasion; 1.8x1.2cm"
[37,] "size of invasion; 1.75x1.3x1.0cm"
[38,] "size of invasion; 2.1x1.8cm"
[39,] "size of invasion; 2.5x2cm"
[40,] "size of invasion; 1.9x1.0cm"
[41,] "size of invasive tumor: 5.7x3.2x2.3cm"
[42,] "size of invasion; 12x3cm"
```

T stage

```
patternx <- "(stage|size[~d]).*T[\\d\\w]{1,3}"
patternx %>% str_match(pathology$report, pattern = .) %>% unlist
```

[,1]	[,2]
[1,] "stage: pT2a"	"stage"
[2,] NA	NA
[3,] "stage: pT2a"	"stage"
[4,] "stage: pT2b"	"stage"
[5,] "stage: pT3"	"stage"
[6,] "stage: pTmi"	"stage"
[7,] "stage: pT2a"	"stage"
[8,] "size; 4.2x4.1cm (T2b"	"size;"
[9,] "stage: pT2b"	"stage"
[10,] "stage: pT2b"	"stage"
[11,] "stage: pT1b"	"stage"
[12,] "stage: pT3"	"stage"
[13,] "stage: pT1b"	"stage"
[14,] "stage: pT3"	"stage"
[15,] "stage: pT2a"	"stage"
[16,] "stage: pT1a"	"stage"
[17,] "stage: pT2a"	"stage"
[18,] "stage: pT1a"	"stage"
[19,] "stage: pT1b"	"stage"
[20,] "stage: pT1b"	"stage"
[21,] NA	NA
[22,] "stage: ypT3"	"stage"
[23,] "stage: pT1b"	"stage"
[24,] "stage: pT2a"	"stage"
[25,] "stage: pT1c"	"stage"
[26,] "stage: pT3"	"stage"
[27,] "stage: pT1b"	"stage"
[28,] "stage: pT3"	"stage"
[29,] "stage: pT1c"	"stage"
[30,] "stage: pT4"	"stage"
[31,] "stage: pT1mi"	"stage"
[32,] "stage: pT2a"	"stage"
[33,] "stage: pT3"	"stage"

[34,]	"stage: pT1c"	"stage"
[35,]	"stage: pT1c"	"stage"
[36,]	"stage: pT1b"	"stage"
[37,]	NA	NA
[38,]	"stage: pT3"	"stage"
[39,]	"stage: pT1c"	"stage"
[40,]	"stage: pT2a"	"stage"
[41,]	"stage: pT3"	"stage"
[42,]	"stage: pT4"	"stage"

N stage

```
patternx <- "stage.*N[\\d\\w]{1,3}"
patternx %>% str_match(pathology$report, pattern = .) %>% unlist
```

	[,1]
[1,]	"stage: pN0"
[2,]	NA
[3,]	NA
[4,]	"stage: pN2"
[5,]	"stage: pN0"
[6,]	"stage: pN0"
[7,]	"stage: pN1"
[8,]	NA
[9,]	"stage: pN0"
[10,]	"stage: pN2"
[11,]	"stage: pN0"
[12,]	"stage: pN0"
[13,]	"stage: pN0"
[14,]	"stage: pN0"
[15,]	"stage: pN1"
[16,]	"stage: pN0"
[17,]	"stage: pN1"
[18,]	"stage: pN0"
[19,]	"stage: pN1"
[20,]	"stage: pN0"
[21,]	NA
[22,]	"stage: ypN0"
[23,]	NA
[24,]	"stage: pN0"
[25,]	"stage: pN0"
[26,]	"stage: pN0"
[27,]	"stage: pN0"
[28,]	"stage: pN1"
[29,]	"stage: pN0"
[30,]	"stage: pN0"
[31,]	NA
[32,]	NA


```
[33,] "stage: pN1"  
[34,] "stage: pN0"  
[35,] "stage: pN2"  
[36,] "stage: pN0"  
[37,] "stage: pN0"  
[38,] "stage: pN1"  
[39,] "stage: pN2"  
[40,] "stage: pN0"  
[41,] "stage: pN2"  
[42,] "stage: pN0"
```